MILITARY SPECIFICATION

HARNESS, PILOTS' SHOULDER SAFETY, STANDARD V TYPE

This specification has been approved by the Bureau of Aeronautics, Department of the Navy.

1. SCOPE AND CLASSIFICATION

1.1 Scope.- This specification establishes the requirements for one type of pilots' shoulder harness.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and drawings of the issue in effect on date of invitation for bids, shall form a part of this specification to the extent specified herein:

2.1.1 Specifications.-

Federal

NN-B-591	Boxes, Folding, Fiberboard, Corrugated, Triple Wall
NN-B-601	Boxes, Wood, Cleated Plywood (For Domestic Shipment)
NN-B-621	Boxes, Wood Nailed and Lock Corner
NN-B-631	Boxes, Wood, Wirebound (Overseas Type)
LLL-B-631	Boxes, Fiber, Corrugated (For Domestic Shipment)
LLL-B-636	Box, Fiber, Solid (For Domestic Shipment)
PPP-B-601	Boxes, Wood, Cleated Plywood

Military

MIL-B-107	Boxes, Wood, Wirebound (Overseas Type)
MIL-B-138	Boxes, Wood, Fiberboard - Lined for Overseas Shipment
MIL-I-6903	Ink, Marking (For Parachutes and Other Textile Items)
MIL-L-10547	Liners, case, Waterproof
JAN-P-103	Packaging and Packing for Overseas Shipment - Boxes; Wood Cleated; Solid Fiberboard
JAN-P-106	Packaging and Packing for Overseas Shipment - Boxes; Wood Nailed
JAN-P-108	Packaging and Packing for Overseas Shipment - Boxes; Fiberboard (V-Board and W-Board), Exterior and Interior

2.1.2 Other Publications.-

Standards

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

2.1.3 Drawings

Military Standard

MS 16068(Aer)	Harness, Pilots' Shoulder Safety, Standard V Type

Naval Aircraft Factory

502302	Harness, Pilots' Shoulder Safety, Standard V Type

FSC 1680

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(When requesting specifications, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications may be obtained upon application to the Commanding Officer, U. S. Naval Air Station, Johnsville, Pennsylvania, Attention: Technical Records Division).

3. REQUIREMENTS

3.1 Material.- Materials shall conform to the specifications indicated on the applicable manufacturing drawings. When materials which are not specifically designated are used, they shall be entirely suitable for the purpose and shall be subject to approval of the Procuring Activity.

3.2 Design and Construction.- The shoulder harness shall be as shown on MS drawing 16068(Aer) and NAF drawing 502302.

3.3 Strength.-

3.3.1 Proof Load.- The shoulder harness shall withstand the 2775 pound tensile proof load described in section 4 without failure or signs of weakening in the stitching. There shall be no appreciable slippage of the webbing through the adjustment loops during application of the load. After removal of the load the hardware shall show no deformation, and the webbing shall show no signs of yarn breakage.

3.3.2 Ultimate Load.- The shoulder harness shall withstand the 3775 pound tensile ultimate load described in section 4. Deformation of fittings or partial failure of webbing or stitching shall not constitute cause for rejection.

3.4 Marking.- Each shoulder harness shall be marked with the manufacturer's name and the MS part number in letters and numerals 1/2 inch high using ink conforming to Specification MIL-I-6903.

3.5 Workmanship.- Because of the emergency nature of the equipment, it is required that all details of workmanship be of the highest quality to insure proper functioning under conditions to which the shoulder harness may be subjected. Unsatisfactory workmanship such as in the form of loose, broken or incomplete stitching and rough, malformed, misaligned or improperly machined fittings shall be considered as defects.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of Tests.- The inspection and testing of the shoulder harness shall be classified as follows:

(a) Preproduction Test.- Preproduction tests are those tests accomplished on samples which are representative of the production item after the award of contract, to determine that the production meets the requirements of this specification.

(b) Acceptance Tests.- Acceptance tests are those tests accomplished on harnesses manufactured and submitted for acceptance under contract.

4.2 Sampling.-

4.2.1 Pre-production Testing.- At the beginning of production under each contract, two assemblies shall be submitted for pre-production testing and shall be furnished by the manufacturer in addition to the quantity specified on the contract or order and without cost to the Government. In addition, samples of webbing (5 yards), thread (1/4 pound), and one each of the hardware used shall be also forwarded to indicate the materials used. Fabrication of any further safety harnesses shall be withheld until samples submitted are pronounced satisfactory by the Procuring Activity. Preproduction testing shall consist of meeting all requirements of this specification. When a contractor is in continuous production of safety harnesses from contract to contract, submission of further pre-production samples on the new contract may be waived at the discretion of the Procuring Activity.

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Samples shall be forwarded to the Naval Aircraft Factory, Naval Air Material Center, Philadelphia 12, Pa., plainly identified by a securely attached durable tag marked with the following information:

Sample for Pre-production Test
Harness, Pilots' Shoulder Safety, Standard V Type
Specification MIL-H-18970 (Aer)
MSI6068 (Aer)
Name of Manufacturer
Submitted by (name) (date) for pre-production test in accordance with Specification MIL-H-18970 (Aer) under Contract Number_____.

4.2.2 Aceptance Tests.- Acceptance tests shall consist of the Inspection Tests (4.3.2) and the proof load test (4.3.3.1). The proof load test shall be conducted on a random sample selected from each inspection lot in accordance with MIL-STD-105, Inspection Level I. However due to the critical nature of the item, the rejection number for all sample sizes shall be one

4.2.2.1 Lot.- An inspection lot shall consist of harnesses made of the same material manufactured under essentially the same conditions and presented for acceptance at the same time.

4.3 Tests and Test Methods.-

4.3.1 Pre-production.- Pre-production tests of safety harnesses shall consist of Inspection, all Acceptance Tests, and the following tests:

4.3.1.1 Ultimate Load.- The shoulder harness shall be assembled to a testing machine as shown in Figure 1 with the left and right straps adjusted to equal lengths. An ultimate load of 3775 pounds shall then be gradually applied to the harness, at a rate of separation of the heads of one inch per minute (under no load). After the specified load has been applied for 30 ± 3 seconds it shall be removed.

4.3.2 Inspection.- Each harness shall be carefully examined to determine conformance with this specification with respect to materials, workmanship, and accuracy of assembly and marking.

4.3.3 Acceptance.- Random samples from each inspection lot shall be subjected to the following tests:

4.3.3.1 Proof Load.- The shoulder harness shall be assembled in the testing machine in the same manner as for Ultimate Load testing. A proof load of 2775 pounds shall then be gradually applied to the harness, at a rate of separation of the heads of one inch per minute (under no load). After the specified load has been applied for 30 ± 3 seconds it shall be removed.

4.4 Retests.- Lots which have been rejected may be reworked or defective items replaced and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original lot shall be furnished the Inspector.

5. PREPARATION FOR DELIVERY

5.1 Application.- The packaging, packing and marking requirements, specified herein apply only to direct purchases by or direct shipments to the Government.

5.2 Packaging.-

5.2.1 Unit Packages.- Unless otherwise specified by the Procuring Activity each safety harness shall be unit packaged within a container conforming to Specification JAN-P-108 or in a 200 pound minimum Mullen test fiberboard box.

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5.3 Packing.- Unless otherwise specified, all items shall be packed for domestic shipment and storage. Shipping containers shall contain the same number of unit packages and shall be uniform in size and snugly packed.

5.3.1 For Domestic Shipment and Storage.- Unless otherwise specified, the harnesses packaged as described in paragraph 5.2 shall be packed in wood-cleated-fiberboard, wood-cleated-plywood, nailed wood, wire-bound wood, corrugated fiber or solid fiber boxes conforming to Specification NN-B-591, NN-B-601, NN-B-621, NN-B-631, LLL-B-631 and LLL-B-636, respectively. The gross weight of wood or wood-cleated boxes shall not exceed 200 pounds. The gross weight of fiberboard boxes shall be subject to the limitation of the box specification.

5.3.2 For Overseas Shipment.- When specified, the safety harnesses packaged as described in paragraph 5.2 shall be packed in wood-cleated solid fiberboard, cleated-plywood, nailed wood, wirebound wood, fiberboard, or fiber-lined wood boxes, conforming to Specifications JAN-P-103, PPP-B-601, JAN-P-106, MIL-B-107, JAN-P-108 and MIL-B-138, respectively. The gross weight of wood or wood cleated boxes shall not exceed 150 pounds; of fiberboard boxes, 70 pounds. The container shall be fitted with a sealed water proof case liner conforming to Type I, Grade B, Class 2 of Specification MIL-L-10547.

5.4 Marking.- Each unit package and exterior shipping container shall be plainly and legibly marked as specified in MIL-STD-129, and shall include the following nomenclature:

Harness, Pilots' Shoulder Safety, Standard V Type

6. NOTES

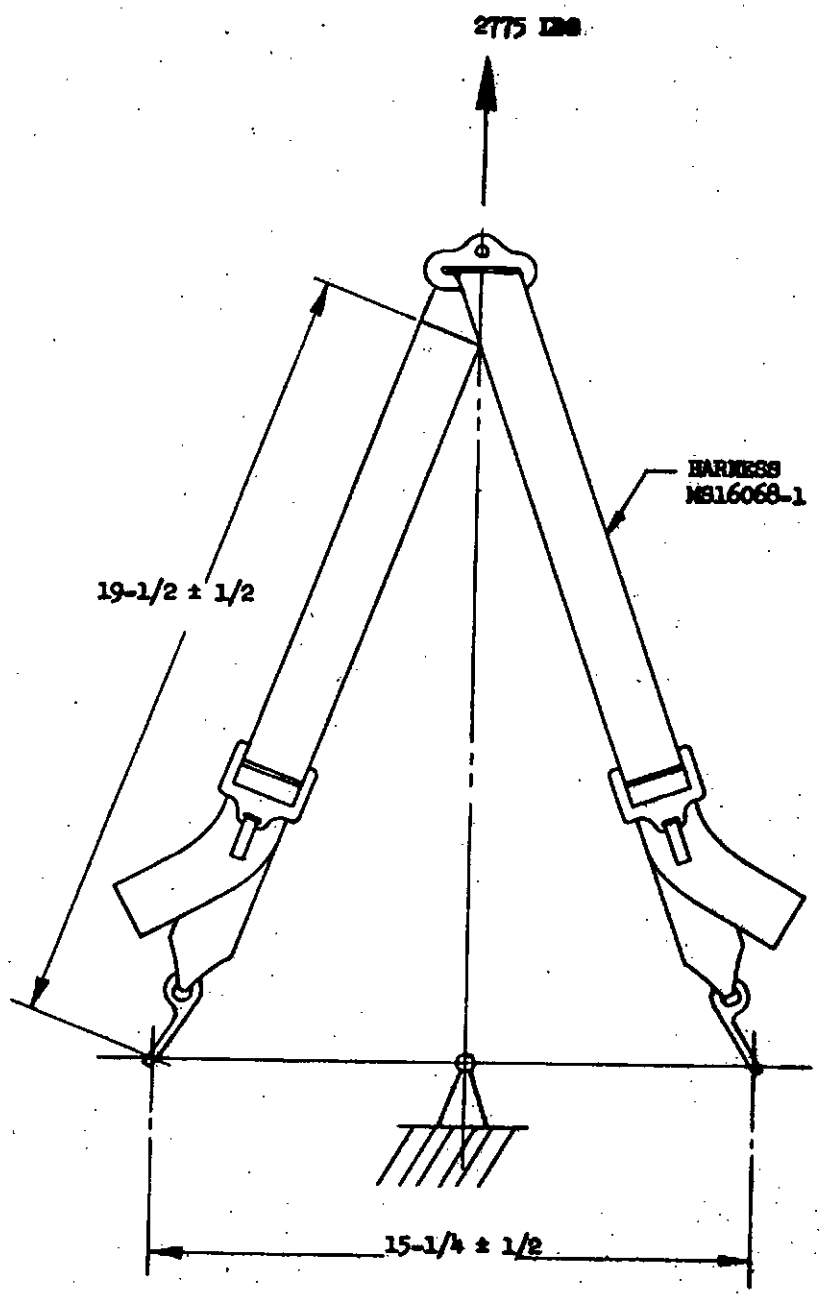
6.1 Use.- Harnesses covered by this specification are intended for use in certain naval aircraft, as safety equipment.

6.2 Ordering Data.- Requisitions, contracts and orders should specify safety harnesses by designating this specification number and the applicable MS part number (see MS 16068(Aer)).

NOTICE: When Government drawings, specifications or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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DIMENSIONS IN INCHES.

FIGURE 1. LOADING CONFIGURATION

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